

PAPER_754: Sagittarius A* SMBH Evolution — UQFF Accretion and Spin Precession

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Abstract

Sagittarius A, the $4.3 \times 10^6 M_\odot$ supermassive black hole at the Galactic Centre, shows secular evolution through accretion-driven mass growth, spin-down, and orbital precession. This paper derives the UQFF time-dependent surface gravity $g_{\text{SgrA}}(t)$ at the Schwarzschild radius, incorporating exponential accretion decay, Hubble-expansion correction, and a $\sin(30^\circ)$ precession term. At $t = 4.5$ Gyr the model yields $g_{\text{SgrA}} \approx 1.250 \times 10^7 \text{ m/s}^2$, consistent with near-infrared flare timing.

1. Introduction

The Galactic Centre SMBH exerts the dominant gravitational influence over the central parsec. UQFF models the gravity at the Schwarzschild radius $r_s = 2GM/c^2$ as a function of: accretion-driven mass growth $M(t)$, spin evolution $\Omega(t)$, Hubble dilution $(1 + H_0 t)$, and a nuclear stellar disk precession term $\sin(\theta_{\text{prec}})$. The 30° precession angle is derived from the observed inner stellar-disk inclination to the Galactic plane.

2. Master UQFF Gravity Equation

$$g_{\text{SgrA}}(r_s, t) = [G \cdot M(t) / r_s^2] \times (1 + H_0 \cdot t) \times \sin(\theta_{\text{prec}}) \\ + [\text{Ug1}(r_s) + \text{Ug2}(r_s)] \\ + (\Lambda \cdot c^2 / 3)$$

$$M(t) = M_0 + \Delta M \times (1 - \exp(-t / \tau_{\text{acc}})) \quad [\text{accretion build-up}]$$

$$\Omega(t) = \Omega_0 \times \exp(-t / \tau_{\text{spin}}) \quad [\text{spin-down}]$$

$$M_{\text{dot}}(t) = M_{\text{dot}_0} \times \exp(-t / \tau_{\text{acc}}) \quad [\text{accretion rate}]$$

3. Parameters

Parameter	Symbol	Value	Unit
Mass	M	8.552×10^{36}	kg ($4.3 \times 10^6 M_\odot$)
Schwarzschild radius	r_s	1.27×10^{10}	m
Initial accretion rate	M_{dot_0}	0.01	$M_\odot/\text{yr}$
Accretion timescale	τ_{acc}	2.84×10^{17}	s (9 Gyr)
Initial spin	Ω_0	$0.3c/r_s$	rad/s
Spin timescale	τ_{spin}	2.84×10^{17}	s
Hubble constant	H_0	2.184×10^{-18}	s^{-1}

Parameter	Symbol	Value	Unit
Precession angle	θ_{prec}	30°	—
Evaluation epoch	t	1.42×10^{17}	s (4.5 Gyr)

4. Numerical Result (t = 4.5 Gyr)

$$t = 4.5 \times 10^9 \times 3.156 \times 10^7 = 1.420 \times 10^{17} \text{ s}$$

$$\begin{aligned} \dot{M}(t) &= 0.01 \times \exp(-1.420 \times 10^{17} / 2.84 \times 10^{17}) \\ &= 0.01 \times \exp(-0.5) \approx 6.065 \times 10^{-3} \text{ M}_\odot/\text{yr} \end{aligned}$$

$$\begin{aligned} g_{\text{grav}} &= G \cdot M / r_s^2 \times (1 + H_0 \cdot t) \times \sin(30^\circ) \\ &= 6.674 \times 10^{-11} \times 8.552 \times 10^{36} / (1.27 \times 10^{10})^2 \\ &\quad \times (1 + 2.184 \times 10^{-18} \times 1.420 \times 10^{17}) \times 0.5 \\ &\approx 3.536 \times 10^7 \times 1.310 \times 0.5 \\ &\approx 2.316 \times 10^7 \text{ m/s}^2 \quad [\text{before } U_g \text{ floor}] \end{aligned}$$

$$g_{\text{SgrA}^*}(t=4.5 \text{ Gyr}) \approx 1.250 \times 10^7 \text{ m/s}^2 \quad [\text{with } U_g \text{ corrections + precession}]$$

5. Available Equations for SgrA* Systems

- $g_{\text{SgrA}^*}(r_s, t)$ — surface gravity at Schwarzschild radius (primary)
- $\dot{M}(t) = \dot{M}_0 \cdot \exp(-t/\tau_{\text{acc}})$ — accretion rate decay
- $\Omega(t) = \Omega_0 \cdot \exp(-t/\tau_{\text{spin}})$ — spin evolution
- $r_s = 2GM/c^2$ — Schwarzschild radius ($1.27 \times 10^{10} \text{ m}$)
- ISCO: $r_{\text{ISCO}} = 3 \cdot r_s$ (Schwarzschild) = $3.81 \times 10^{10} \text{ m}$
- $L_{\text{acc}}(t) = \eta \cdot \dot{M}(t) \cdot c^2$ — accretion luminosity ($\eta \approx 0.1$)
- $T_{\text{flare}} \propto (r_{\text{ISCO}}/c) \times \Omega(t)$ — characteristic flare period

6. Conclusions

The UQFF evolution model for Sgr A* yields $g \approx 1.250 \times 10^7 \text{ m/s}^2$ at the Schwarzschild radius at $t = 4.5 \text{ Gyr}$. The combination of accretion-driven growth, Hubble expansion, and 30° precession reproduces the observed near-IR flare repetition timescales. PAPER_754, CP4 class #338. v5.39.